

Piyush Kumar Singh

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PROFESSIONAL SUMMARY

Backend-strong Full Stack Engineer with production experience in Node.js, distributed systems, and real-time APIs. Ranked #32 among 3,414+ global contributors (Top 1%) in GSSoC 2025 across 10+ production repositories. Open to backend and full stack SDE internship roles.

TECHNICAL SKILLS

Languages	JavaScript (ES2022+), TypeScript, Python, C++, Java, SQL
Frontend	React.js, Next.js, HTML5, Tailwind CSS, REST API integration
Backend	Node.js, Express.js, FastAPI, REST APIs, Microservices, JWT, RBAC, Rate Limiting, WebSocket/Socket.io, SSE, AES-256, OTP Auth, Refresh Token Rotation
Databases	MongoDB (indexing, aggregation), PostgreSQL, MySQL, Redis — Schema Design, Query Optimisation
DevOps	Docker, Git, GitHub Actions, CI/CD, Render, Vercel, Postman, Linux
Core CS	Data Structures & Algorithms, OOP, DBMS, System Design, OS, Computer Networks

EXPERIENCE

LLM Post-Training Intern | Ethara AI

On-site | Dec. 2025 – Jan. 2026

- Evaluated 10,000+ LLM responses daily against structured rubrics; surfaced recurring failure patterns in multi-step reasoning, factual grounding, and instruction-following that fed directly into model improvement cycles.
- Designed and standardised a prompt–response evaluation framework adopted team-wide, reducing inter-reviewer inconsistency and compressing feedback turnaround by ~30%.
- Performed root-cause error analysis on edge-case outputs, producing annotated correction reports that helped prioritise high-impact fine-tuning targets.

Open Source Engineer — Top 1% Global Contributor | GirlScript Summer of Code

Remote | Aug. 2025 – Nov. 2025

- Merged 48+ PRs across 10+ production repos contributing 5,000+ lines of peer-reviewed code — ranked #32 / 3,414+ (Top 1% globally).
- Designed and shipped a JWT + RBAC auth system for 100+ concurrent users, replacing a session-cookie approach with known security gaps.
- Optimised 6 REST API endpoints and refactored MongoDB query plans (compound indexes, N+1 elimination), cutting response time by 40%.

PROJECTS

MEDIQUEUE [GitHub](#) | [Live](#) | Node.js · Express.js · React.js · MongoDB · Redis · Socket.io · JWT · Docker · AES-256 · GitHub Actions

- Architected 100+ REST APIs across 13 route modules and 9 MongoDB models; compound indexing cut average DB query latency by 55% under production-like load.
- Built an AES-256-CBC encrypted medical records vault with field-level RBAC, securing 200+ patient profiles — passed internal pen-test review.
- Engineered real-time queue via Socket.io for 50+ concurrent users at sub-200 ms latency; Redis rate limiting capped at 100 req/min per client.
- Set up GitHub Actions CI/CD pipeline — automated test, lint, and Docker build on every PR, eliminating manual deployments and enforcing consistent release quality.

Travira — Microservices Safety Platform [GitHub](#) | [Live](#) | Node.js · Express · Python FastAPI · React.js · MongoDB · WebSocket · Docker

- Architected 5 independently deployable microservices with 25+ REST APIs; shared auth middleware cut cross-service integration overhead by 60%.
- Built a Python FastAPI AI scoring microservice with real-time WebSocket streaming, delivering threat scores to 30+ concurrent clients in under 300 ms.
- Deployed all 5 services to Render with zero-downtime strategy; unified API gateway served 3 client platforms (Android, React web, admin panel) from a single entry point.

EDUCATION

B.Tech — Information Technology | Madhav Institute of Technology and Science, Gwalior

Aug. 2023 – Aug. 2027

- CGPA: 8.14 / 10 · Relevant Coursework: DSA, DBMS, OOP, System Design, OS, Computer Networks

ACHIEVEMENTS & CERTIFICATIONS

- 98.55 Percentile among 500,000+ participants — Naukri Campus Young Turks 2025.
- Ranked #32 / 3,414+ contributors (Top 1%) — GirlScript Summer of Code 2025; 48+ merged PRs, 5,000+ LOC.
- Solved 499+ DSA problems (140E / 292M / 67H) with 800+ GitHub contributions.
- Oracle Cloud Infrastructure 2025 Certified: Generative AI Professional + AI Foundations Associate.
- Postman API Fundamentals Student Expert (Sep 2025).